Model Overview:
The model predicts pitching performance for the next season using statistics from the previous season. The main goal of the model is to get a better understanding of how previous season performance correlates to next season's performance in order to aid offseason acquisitions.

Training Data:
The model was training on data from 2018 to 2022. Where statistics from 2018 would be used to predict the players performance for 2019. Data was filtered so that the minimum innings pitched was 50 innings and we used FIP- as our target variable and set the positive outcome to pitchers with FIP- under 100 which means the pitchers FIP was better than league average. The validation set consisted of data from 2023 and the model was tested on data from 2024.

Features:
There were 9 features which were included in the model:

1. K/9+: Strike outs per nine innings on a scale where 100 is the league average. This provides the model with a good gauge into how dominant a pitcher is compared to league average.
2. BB/9+: Base on bases on a scale where 100 is the league average. This provides the model with a good understanding of how many free passes are given up. Very important because this is one of the main things pitchers can actually control
3. HR/9+: Home runs on a scale where 100 is the league average. How many home runs pitchers are giving up per 9 innings compared to league average.
4. GB%: Pitchers ground ball %. Good estimator since ground balls usually result in better outcomes than fly balls given the chance of a double play and the fact that ground balls are usually held to at most a single.
5. LOB%: Left on base percentage allows the model to understand how good a pitcher is at getting out of jams. This could be seen as more of a luck statistic but the model could pick up on the fact that lower ERA but higher LOB% leaves a lot of opportunity for a worse season.
6. ERA: Pitchers earned run average

Model Performance:
The model performance on the test set are as follows:

- AUC: 0.698
- Class 1 Precision: 0.71
- Class 1 Recall: 0.63
- Class 1 F1: 0.66
- Class 0 Precision: 0.55
- Class 0 Recall: 0.64
- Class 0 F1: 0.59

These metrics mean the model was able to extract a relationship between the previous seasons statistics and next seasons FIP-. The strength of the relationship is weak which leads to slight inaccuracies in predictions. The model performs best when predicting players who are likely to perform well next season.

5. How to Interpret Output:

The model output should be interpreted as a likelihood where the value represents the probability that the model thinks a pitcher will have a FIP- under 100. For example if the model outputs .75 for a pitcher that means the model believes there is a 75% chance the player will pitch well next season. This means the model output should strictly be used as a probability and it's highly suggested to only take action on pitchers the model is most confident on the recommended range is .75 for positive players where the model is 85% accurate on the test set which totaled to 33 pitchers. The most important features from the model where K/9+, HR/9+ and GB% in order of importance.

6. Know Limitations:

- The model put a lot of emphasis on a pitcher's K/9+ so pitchers other statistics such as walk rate should be checked before making decisions.
- The model was only trained on pitchers with over 50 innings pitched so the model should not be used on pitchers below that threshold.
- Some features within the model like LOB%+ and HR/9+ are known to be "luck" features and could encode false signals.
- The model was specifically designed to predict the FIP- likelihood which mainly encodes walks, homeruns, and strikeouts so it will not work well when looking for specific types of pitchers.
- The data within the model largely consisted of starting pitchers so use on relievers or closers is not advised.

7. Out-of-Scope Uses:

- The model should not be used on pitchers returning from significant injury because it was solely trained on previous year data.
- The model should not be used with minor league data as relationships may differ between levels.
- The model should not be used as the sole reason why decisions are made.
- The model should not be used on pitchers which have been traded multiple times within a season as those stats were not encoded and can have a sizable impact on player performance.

8. Recommendations for Use:

The model should mainly serve as a guide for understanding which pitchers who might have struggled within a specific season are likely to pitch better the next season. The model should be used to aid decision making on which pitchers can be signed at below their market price. When using the model human judgment should be used to look into the underlying statistics to ensure the predictions make sense in baseball terms.